

Yash Agrawal

Software Engineer II

✉ yashagrawal070@gmail.com | 📞 +91 9822945017 | 📍 Pune, India
🌐 linkedin.com/in/agrawal-yash1 | 📄 github.com/YashAgrawal1

Professional Summary

Backend engineer with 3.5+ years specializing in search and data-intensive systems at 300M+ record scale. Designed and owned Elasticsearch index architecture, analyzer chains, and zero-downtime reindexing pipelines across B2B data and government hiring platforms, alongside deep PostgreSQL and Ruby on Rails performance work. Currently own end-to-end technical direction for two production platforms at American Technology Consulting, where I set backend design standards and built the internal LLM engineering framework on Anthropic API + MCP.

Technical Skills

Backend: Ruby on Rails, ActiveRecord, Sidekiq, Kafka, REST APIs, Service Objects, GraphQL

Search & Data: Elasticsearch, PostgreSQL, MySQL, MongoDB, Redis

Languages: Ruby, SQL, JavaScript, TypeScript, Java

AI Engineering: Anthropic API, Model Context Protocol (MCP), LLM tool integration

Frontend: React.js, HTML, CSS

Cloud & Infra: AWS (EC2, S3, RDS, SQS), Docker

Tooling & Monitoring: Git, GitHub Actions, RSpec, Postman, Sentry, CloudWatch

Work Experience

American Technology Consulting, Hyderabad

Feb 2026 – Present

Software Engineer II

- Built an internal **AI-powered engineering assistant framework** using **Anthropic API + MCP**, integrating **Jira and Figma** to auto-generate design/AC gap analysis, edge cases, engineering design drafts, and test cases from a Jira Epic ID — reducing ramp-up time on new stories across engineering teams.
- Designed the **Elasticsearch candidate-search layer** for a government ATS over **10M+ applicant records** — custom analyzer chain and field-level boosting for multi-field queries at **70ms p95**.
- Cut p95 on student- and teacher-facing dashboards from **2s to 400ms** by eliminating **N+1 patterns (70 queries per request down to 8)**, redesigning PostgreSQL composite indexes, and extracting hot paths into **service objects**.
- Built and hardened **Sidekiq pipelines processing 100K+ operations** — bulk applicant imports, enrichment, and notifications — with **idempotency and retry-with-backoff**, and dead-letter handling so multi-hour import jobs survive partial failure.
- Set **backend design standards** across both products: introduced **service-object and ES mapping-review conventions**, ran design reviews, and brought **3 engineers** to independent feature ownership on complex Rails + Elasticsearch work.

Data Axle Solutions, Pune

Jul 2023 – Feb 2026

Software Engineer

- Owned the **ingestion and indexing pipeline feeding 300M+ B2B records** into Elasticsearch — incremental updates plus full reindexes behind **zero-downtime alias swaps**, taking full reindex from a maintenance-window operation to a live one at **11h runtime**.
- Redesigned index structure, analyzers, and relevance scoring across **300M+ records**, cutting multi-field query **p95 from 1.5s to 200ms** on high-cardinality fields.
- Profiled **ActiveRecord query plans** and restructured **hot-path PostgreSQL operations** across data-heavy workflows, reducing **p95 on the critical path by 35%**.
- Shipped **20+ end-to-end features** on the core **Rails + PostgreSQL** platform, from schema design through **React.js** integration.
- Operated as **senior IC** in Agile delivery: guided **2 junior developers**, drove code-review standards, standardized backend patterns adopted across the team.

Frontend Developer Intern

Feb 2023 - June 2023

- Built Rails API endpoints, ActiveRecord models, and backend logic alongside React.js work — converted to full-time backend engineering on the strength of backend contributions.
- Developed responsive React.js interfaces consuming REST APIs, improving render performance through lazy loading and component-level optimization.

Education

MIT Academy of Engineering, Alandi, Pune

B.Tech in Computer Science

2019-2023

CGPA: 8.76/10